

SMART REAL-TIME AGE AND EMOTION-BASED ADVERTISING IN MALLS USING CUSTOM-BUILT MODELS

Project ID: 25-26J-091

Project Proposal Report

Wasala Dharmakeerthige Ujali Imalika Samaraweera

B.Sc. (Special Hons) in Information Technology – Data Science

Department of Computer Science

Sri Lanka Institute of Information Technology

Sri Lanka

August 2025

**REAL-TIME EMOTION DETECTION WITH OCCLUSION
HANDLING AND DOMAIN-SPECIFIC CLASSIFICATION FOR
ADAPTIVE ADS AND SECURITY ALERTS**

Project ID: 25-26J-091

Project Proposal Report

Wasala Dharmakeerthige Ujali Imalika Samaraweera

Supervisor: Dr.Bhagya Nathali Silva

B.Sc. (Special Hons) in Information Technology – Data Science

Department of Computer Science

Sri Lanka Institute of Information Technology

Sri Lanka

August 2025

Declaration

Declaration

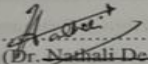
We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student ID	Signature
Samaraweera W.D.U.I.	IT22258526	

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the supervisor:

Date


(Dr. Nathali De Silva) Bhagya Nathani Silan

25/08/2025

Abstract

This project develop real time emotion detection system capable of operating effectively under facial occlusions. It classifies domain-specific emotions relevant to retail and security contexts *Curious*, *Ignoring*, and *Suspicious* and generates two outputs: (1) aggregated crowd emotion metadata for adaptive advertising and (2) security alert when suspicious emotional patterns persist. To ensure privacy, all processing is performed on edge devices without storing images or personal identifiers. The system is trained using a combination of the Sri Lankan facial datasets , occlusion focused datasets for contextual domain specific emotion label. The model incorporate an occlusion aware feature extraction branch focusing on eye region and head-pose analysis, coupled with a temporal module for detecting suspicious trends. This research presents a novel, mask-robust, privacy-conscious emotion recognition pipeline optimized for real time deployment, integrating both adaptive advertising and proactive security monitoring in shopping malls.

Keywords: Emotion Detection, Occlusion Handling, Domain-Specific Emotions, Adaptive Advertising, Deep Learning.

Table of Content

Declaration	I
Abstract.....	II
Table of Content	III
List of Figures.....	IV
List of Tables.....	IV
List of Abbreviations.....	IV
Introduction	1
Background & Literature survey	2
Research Gap.....	5
Research Problem	7
Objectives	8
Main Objectives	8
Specific Objectives	8
Methodology	9
Understanding the key pillars of the research domain	9
Application of technologies	10
Mathematical Equations for the Proposed System	11
System Architecture Diagram	15
Proposed Project Execution	16
Tasks and Sub tasks Identified to Achieve the Objectives	17
Data Requirements and Collection Methods	17
Anticipated Conclusion and Real-World Application	18
Commercialization Aspects of the product	19
Work Breakdown Structure	21
Gantt Chart	22
Use Case Diagram	23
Description of Personal and Facilities	24
Project requirements	25
System Requirements.....	25
User Requirements.....	27
Expected test cases (Tentative)	28
Budget and Budget Justification	30
References.....	31

List of Figures

<i>Figure 1 : Mathematical Model Flow Diagram</i>	<i>13</i>
<i>Figure 2 : Facial Expression Recognitioon</i>	<i>14</i>
<i>Figure 3 : Raspberry Pi-based real-time emotion architecture</i>	<i>14</i>
<i>Figure 4 :System Architecture Diagram of the component</i>	<i>15</i>
<i>Figure 5 :Work breakdown chart</i>	<i>21</i>
<i>Figure 6 :Gantt Chart</i>	<i>22</i>
<i>Figure 7 : Use case Diagram of the component</i>	<i>23</i>

List of Tables

<i>Table 1 :Comparison between existing system</i>	<i>5</i>
<i>Table 2 : Expected test cases</i>	<i>28</i>
<i>Table 3 : Budget Estimation</i>	<i>30</i>

List of Abbreviations

Term	Definition
CNN	Convolutional Neural Network
API	Application Programming Interface
FER	Facial Emotion Recognition
AI	Artificial Intelligence

Introduction

Shopping malls in Sri Lanka rapidly use digital aggregation and camera surveillance to enhance experience and security of customers. However, many advertising systems remain inactive, displaying content on fixed schedules without considering the audience's presence, demographics, or engagement. These results in a gap between viewers and displayed content and reducing advertising effectiveness and customer satisfaction. Security observation often relies on manual observation or simple motion detection which can miss delicate behavioral cues indicating potential risks or suspicious activity.

Recent progression in video analysis and computer vision technologies enable the development of real time systems that analyze video feeds to provide actionable insights for marketing and security. By assessing demographic data, emotional states, and behavioral engagement, digital collection can adapt dynamically to the current audience, while security teams receive timely alert based on observed behaviors.

Real world implementation present challenges like facial occlusions caused by masks, varying lighting, low-resolution footage, and cultural differences in expressions and behavior. Privacy concerns further require that processing happen locally, without storing personally recognizable information.

This research proposes an integrated, real time system made-to-order for Sri Lankan shopping malls. The system consist of four main components: (1) age and gender classification to segment visitors into demographic groups; (2) emotion classification to recognise domain-specific emotions relevant to advertising and security (3) engagement detection through pose estimation to measure individual engagement without relying on facial data; and (4) advertisement personalize engine that dynamically selects appropriate ads based on aggregated real time audience data.

By investing culturally relevant datasets and advanced video analysis techniques, and deploying on edge devices to preserve privacy, the system target to enhance customer engagement, optimize advertisement relevance and improve security monitoring within retail environments.

Background & Literature survey

1. Overall Background of the Research Domain

In Sri Lankan shopping malls, digital display and cameras surveillance are prevalent, targeting to enhance customer experience and security. However, these systems often operate in isolation, with advertisements displayed on fixed schedules no matter of the audience's current emotional state. This lack of responsiveness can lead to disengagement and reduced advertisement effectiveness. Similarly, security monitoring frequently relies on manual observation or basic motion detection, which may overlook harmful behavioral evidence indicative of potential risks or suspicious activities.

Emotion detection from video stream offers a solution by providing real-time insights into the emotional form of individuals. By analyze facial expressions, body language, and contextual evidence, systems can infer emotions, for example; happiness, sadness, anger, neutrality, or domain specific states such as suspicion and ignoring. This capability enable adaptive advertising and proactive security monitoring, enhancing customer engagement and customer safety.

2. Relevant Studies and Techniques to be Mastered

Implementing effective emotion detection system in public spaces need proficiency in several major areas;

- **Occlusion-Robust Emotion Recognition:** Traditional emotion detection systems may struggle with occlusions caused by face masks. To address these, advance models that focus on analyzing visible facial region, like the eyes and upper face, are essential. Techniques such as gaze tracking and head pose estimation can supplement facial analysis to improve accuracy under occlusions. [1]
- **Domain-Specific Emotion Classification:** While basic emotions like happy, sad, and angry are widely studied, recognizing context specific emotions such as suspicion or ignoring is important for application in sales environments. Developing model that can accurate classifying these nuanced emotions need curated datasets and context aware training methodologies..

- **Privacy Preserving Processing:** Given sensitivity of video data, it imperative that emotion detection systems operate without storing or turning personally identifiable information.. Developing edge computing solutions ensures that data processing occur locally preserving user privacy and complying with data protection regulations,.

3. Current State of the Art

Several existing emotion detection platforms like Affectiva and Microsoft's Emotion API are have demonstrated high accuracy under controlled lab conditions. Affectiva, for instance, has trained it models on millions of real world videos from diverse environments using deep learning and computer vision techniques. However, comprehensive bench marking studies reveal that commercial systems, including Amazon Rekognition, Azure Face API, and Affectiva, see accuracy drop significantly when applied to unconstrained inputs like CCTV footage taken from varied angles and contexts . [2]

Moreover, accuracy declines sharply when faces are partially occluded by masks or other coverings a common scenario in common places. A recent survey notes that most facial expression recognition research remains lab based, with low focus on partial occlusion or real world complexity .

Another limitation is restricted emotion set these systems are trained on. They typically detect basic emotions happy, sad, angry, neutral but lack the capability to recognize domain specific emotions such as suspicion or ignoring behaviors, which are critical for applications in advertising targeting and security monitoring [3] . Moreover, cultural differences in emotional expression remain largely unaddressed in commercial systems, increasing concerns about generalization and fairness in diverse regions like Sri Lanka . [4]

Recent advancements have led to the development of more burly models capable of detecting emotions under occlusions in diverse cultural contexts. For example, research has shown that analyzing the upper face and utilizing regard tracking can improve emotion recognition accuracy when the lower face is unclear. Additionally, creation of culturally diverse datasets [5] has enhanced the generalization of emotion detection models across different populations..

4. Previous Approaches and How This Research Differs

Traditional retail and security systems focus on demographics or motion detection often disregard real time emotional states. Existing emotion detection methods depend on facial features, can be obscured by masks, reducing accuracy. Recent studies highlight occlusion robust approaches using CNNs and ensemble models to maintain detection even under uncomplete face coverage. [6]

This research develops a real time, occlusion robust emotion detection system that classifies basic and domain specific emotions, it delivers dual outputs for advertising and security, and ensures privacy by avoiding image storage.

Unlike traditional systems, the novelty lies in its unified design that combines three unique features:

- **Real time Emotion Detection under Occlusion** – The system maintain the accuracy even when faces are partially covered with masks, caps, or sunglasses, ensuring reliability in real world shopping mall environments.
- **Intelligent Dual Output Filtering** – Predicted emotions are intelligently filtered into two action paths: (1) dynamic advertisement control based on aggregated emotional states, (2) instant real time alerts for suspicious or ignoring behaviors, using a rule based classification technique.
- **Privacy Aware Edge Deployment** – All processing occurs locally on edge devices, ensuring that no personal images are stored, addressing critical privacy and regulatory concerns.

This integrated approach not only advances technical capability (robust, dual output detection under occlusion) but also addresses functional and ethical needs, making it highly novel and directly applicable in real world marketing and security settings.

Research Gap

Despite advances in emotion recognition, there is a significant gap in applying real time facial emotion detection for adaptive advertising and security in shopping malls. Most of existing systems focus on basic emotions (happy, sad, angry, neutral) and ignore domain specific emotions like curiosity, suspicion, ignoring, which are important for targeted ads and early security alerts.. Many emotion detection models fail too when faces are partially occluded by masks, a common scenario in public places. While some research convey occlusion or domain specific emotions individually; very few integrate in a real world pipeline. And also, current solutions rarely provide dual outputs for business applications one for personalized advertising and another for security alert when ensuring privacy by processing data on edge devices without storing images.

This project fulfill this gap by designing an occlusion robust, domain-specific emotion detection system that delivers two actionable outputs (ads and alerts) in real time and preserves user privacy. It leverages culturally relevant datasets, combines masked and unmasked training data, implements edge deployment, offering a unique, production ready solutions are not addressed in prior research.

Table 1: Comparison between existing system

Aspect/Feature	Existing Systems (Research)	Proposed Application Component
Domain-Specific Emotion Classes	Most FER systems detect primary emotions only (happy, sad, angry, neutral). EMOTIC give domain labels for research but lacks integrated production pipelines. [7]	The planned system detect basic and domain-specific emotions (curious, suspicious, ignoring) to improve advertising relevance and security alerts.

Occlusion Handling(Masks)	Existing methods handle occlusion in controlled labs, aim on reconstruction, or target basic emotions only. Accuracy drops in real world crowded situations. [8]	Occlusion robust detection using masked and without masked datasets; works reliably in real-world shopping malls..
Dual-Purpose Outputs (Ads + Security)	Systems mainly focus either on marketing (ads) or security monitoring, rarely, no real time and dual output pipelines exist. [9]	Single integrated pipeline outputs: 1) aggregated emotion data for adaptive advertisement 2) alerts for suspicious behaviors in real time.
Privacy & Edge Deployment	Most research processes and save images in the cloud, raising privacy concerns. A few solutions implement fully edge based real time inference. [10]	Fully edge deployed system that processes video locally and does not store images, make sure privacy by design.

Summary of the Research Gap

The table highlight main gaps between existing systems and the proposed application. The major innovations in the project lie in the development of a real time, occlusion robust emotion detection system that identify basic and domain-specific emotions (curious, suspicious). Unlike existing solutions; it integrates dual purpose output for adaptive advertising and security alert while preserving user privacy through edge deployment, These advancements make the proposed system distinct from previous research and commercial offerings, addressing the unique challenges of understanding audience emotions in dynamic, real world retail environments. By combining culturally relevant datasets; occlusion handling; and real time processing; the system ensures more effective advertising, timely security engagement and ethical data usage filling a important gap in current retail and security technologies.

Research Problem

Current digital advertising and security systems in Sri Lankan shopping malls operate with significant limitations. Advertisements are predominantly static or follow fixed schedules, without adapting to the real-time presence, demographics, or behavioral engagement of the audience. This lack of contextual relevance often leads to ineffective marketing, decreased viewer attention, and lost revenue opportunities for retailers. Simultaneously, mall security relies heavily on manual surveillance and basic motion detection techniques, which are insufficient for early detection of suspicious activities or subtle emotional cues indicative of potential threats.

The advent of artificial intelligence and computer vision offers the potential to revolutionize these systems by enabling real-time, data-driven decision-making based on live video analytics. However, deploying such intelligent system in real world retail environment faces several challenges. First facial occlusions caused by face masks, caps or other coverings introduced especially during and after the COVID 19 pandemic [11], back traditional facial recognition and emotion detection models, leading to unreliable or incomplete data. Second, variations in lighting, camera angle and video quality further degrade detection accuracy [12]. Third, cultural variations in emotional expression necessitate training models on region specific datasets to improve relevance and accuracy. [13] Moreover, privacy concerns impose strict constraints on data collection and storage. Systems must process sensitive video data locally on edge devices avoiding image retention or transmission to safeguard individual identities.

Some other challenge lies in accurately measuring viewer engagement beyond facial expression analysis. Traditionally crowd counting methods fail to capture nuanced behavior like body orientation, stillness and dwell time, which are crucial indicators of engagement in retail space. There is clear need for a robust, privacy conscious system capable of analyzing body pose, behavior without relying on facial data. Hence, this research addresses the problem of designing and implementing a comprehensive real time system for Sri Lankan shopping malls that overcomes these challenges by combining occlusion aware emotion recognition, age classification, pose-based engagement detection, and dynamic advertisement personalization all while respecting privacy and cultural context. This integrated solution aims to enhance advertisement effectiveness and security responsiveness in shopping malls.

Objectives

Main Objectives

The primary objective of this proposed research is to develop a real time emotion detection system capable of recognizing both basic and domain specific emotions (suspicious, ignoring) from live video feeds, even under facial occlusions like masks. The system aim to generate actionable outputs for two distinct applications; adaptive advertising and security alerting, while ensuring privacy through edge based processing.

Specific Objectives

- **Real Time Emotion Detection:**

Developing deep learning models; to accurately detect and classify emotions including happiness, sadness, anger, neutral,, curiosity, suspicious, and ignoring [14] . The system will process live video streams from camera to assess the emotional state of individuals in malls.

- **Occlusion Handling:**

The system is designing to handle common real world occlusions like masks or partial face visibility. Models will leverage eye region ,head pose features to maintain advanced accuracy in detecting basic and domain specific emotions. [15]

- **Dual-Purpose Output Generation:**

Developing mechanism to filter detected emotions into two outputs. Aggregate crowd emotion data for adaptive advertisement selection. Alerts are for security teams when detected suspicious emotional patterns repeatedly.

- **Privacy-Preserving Edge Deployment:**

Ensure all processing occurs locally on edge devices without storing images or personal data. Only anonymize emotion labels will be used for decision-making.

Methodology

Understanding the key pillars of the research domain

In this project, the key pillars include Facial Emotion Recognition (FER), Occlusion/Mask Robustness and Dual Output Domain Specific Emotion Process, Each pillar is critical for achieving the main objective of develop real time system that enhance targeted advertising and improves public safety through integrated security alerts.

- **Facial Emotion Recognition (FER) –**

Specialized Domain: Image Processing and Machine Learning

The system detect faces from live video feeds, extract key facial features and classifies them using a CNN based model into basic emotions (happy, sad, angry, neutral) and domain specific emotions (curious, suspicious, ignoring, bored). Unlike conventional system focused on generic emotions; this model is tuned for contextual relevance to guide for targeted advertising and security monitoring.

- **Occlusion and Mask Robustness –**

Specialized Domain: Image Processing and Data Science

To handle real world challenges like masks, caps or sunglasses, the system includes a mask aware detection module. It identify occluded faces and routes them to a secondary classifier trained on masked datasets, By combining unmasked datasets (Sri Lankan Emotion Data set) with masked datasets it maintains high accuracy even in crowded, partially occluded environments.

- **Dual Output Domain-Specific Emotion Mapping –**

Specialized Domain: Machine Learning and Decision Systems

Emotions that were predicted are filtered into two real time outputs (1) Advertising Path: aggregates crowd emotions to adapt displayed ads; and (2) Security Path: flags suspicious or curious avoidant emotions to trigger instant security alerts. This unified dual output design cuts costs and improve efficiency by removing the need for separate systems and streamlining operations and optimizing resource use effectively.

Application of technologies

This research integrates computer vision, machine learning, and edge deployment to enable robust, real time emotion detection in shopping malls.

1. Facial Emotion Recognition (FER) – Machine Learning & Computer Vision Technologies

Algorithms: Convolutional Neural Networks (CNNs), transfer learning models (ResNet, MobileNet, VGGFace) will be used for real time facial emotion recognition. Rule based filtering will be applied to map detected emotions into advertising and security outputs.

Tools and Libraries: Frameworks such as TensorFlow, PyTorch, OpenCV, and Scikit-learn will support model development, training, and deployment. [16]

2. Occlusion and Mask Robustness – Image Processing & Data Science

Standard FER fails under masks or occlusions. Occlusion aware CNNs with masked datasets improve robustness. A hybrid approach combining Sri Lankan Emotion Dataset with masked datasets ensures accuracy in real world mall environments. [17]

3. Dual-Output Emotion Mapping – Machine Learning & Decision Systems

Advertising Path: Real-time analytics engines (e.g, Apache Kafka) with Python based rule engines will aggregate crowd emotions for adaptive ad delivery.

Security Path: Rule based classifiers (Scikit-learn) with Flask-SocketIO or Node-RED will trigger live alerts for suspicious or ignoring behaviors. [18]

By combining CNN based FER with transfer learning, mask aware classification, rule based dual output filtering, and edge deployment the proposed system ensures real time, accurate and privacy conscious performance. This integrated technological selection differentiates it from existing one dimensional solutions, proving novelty and practical applicability.

Mathematical Equations for the Proposed System

1. Face Detection & Feature Extraction (Convolution Operation)

Facial features are extracted from live video frames using Convolutional Neural Networks (CNNs). Each convolutional layer applies filters to detect patterns such as eyes, mouth, and facial expressions:

$$f_{i,j}^{(l)} = \sigma \left(\sum_{m,n} w_{m,n}^{(l)} \cdot x_{i+m,j+n}^{(l-1)} + b^{(l)} \right)$$

$f_{i,j}^{(l)}$ - output feature map at layer l

$w_{m,n}^{(l)}$ - convolutional filter weights

$x^{(l-1)}$ - input from previous layer

$b^{(l)}$ - bias term

σ - activation function (ReLU)

2. Emotion Classification (Softmax Layer)

The final CNN layer outputs probabilities for all possible emotion (basic + domain-specific):

$$P(y = k | x) = \frac{e^{z_k}}{\sum_{j=1}^K e^{z_j}}$$

$P(y = k|x)$ - probability of emotion class k

z_k - logit score for class k

K - total number of emotion classes

3. Occlusion Handling (Fusion for Masked & Unmasked Faces)

To improve robustness against occlusions (masks, caps), a fusion strategy combines predictions from unmasked and masked classifiers:

$$E = \alpha \cdot E_{\text{unmasked}} + (1 - \alpha) \cdot E_{\text{masked}}$$

E - final predicted emotion

E_{unmasked} - output of model trained on full-face data

E_{masked} - output of model trained on masked data

α - confidence weight (based on occlusion detection)

4. Crowd Emotion Aggregation

For multiple people detected in the frame, the system computes the dominating crowd emotion every few seconds:

$$E_{\text{crowd}} = \arg \max_{c \in C} \left(\frac{1}{N} \sum_{i=1}^N P(e_i = c) \right)$$

E_{crowd} - aggregated emotion of the crowd

N - number of faces detected

$P(e_i = c)$ - probability that person i shows emotion c

C - set of all possible emotions

5. Rule-Based Dual Output Filtering

The detected emotions are mapped into two well-defined outputs: advertising or security.

$$O = \begin{cases} O_{\text{Ad}}, & \text{if } E \in \{\text{happy, neutral, bored}\} \\ O_{\text{Sec}}, & \text{if } E \in \{\text{suspicious, ignoring}\} \end{cases}$$

O_{Ad} - advertising control output

O_{Sec} - security alert output

6. Model Training Objective (Cross-Entropy Loss)

The CNN is trained using categorical cross-entropy loss:

$$\mathcal{L} = - \sum_{i=1}^N \sum_{k=1}^K y_{i,k} \log \hat{y}_{i,k}$$

$y_{i,k}$ - true label (1 if sample i belongs to class k , else 0)

$\hat{y}_{i,k}$ - predicted probability for class k

N - number of samples

K - number of emotion classes

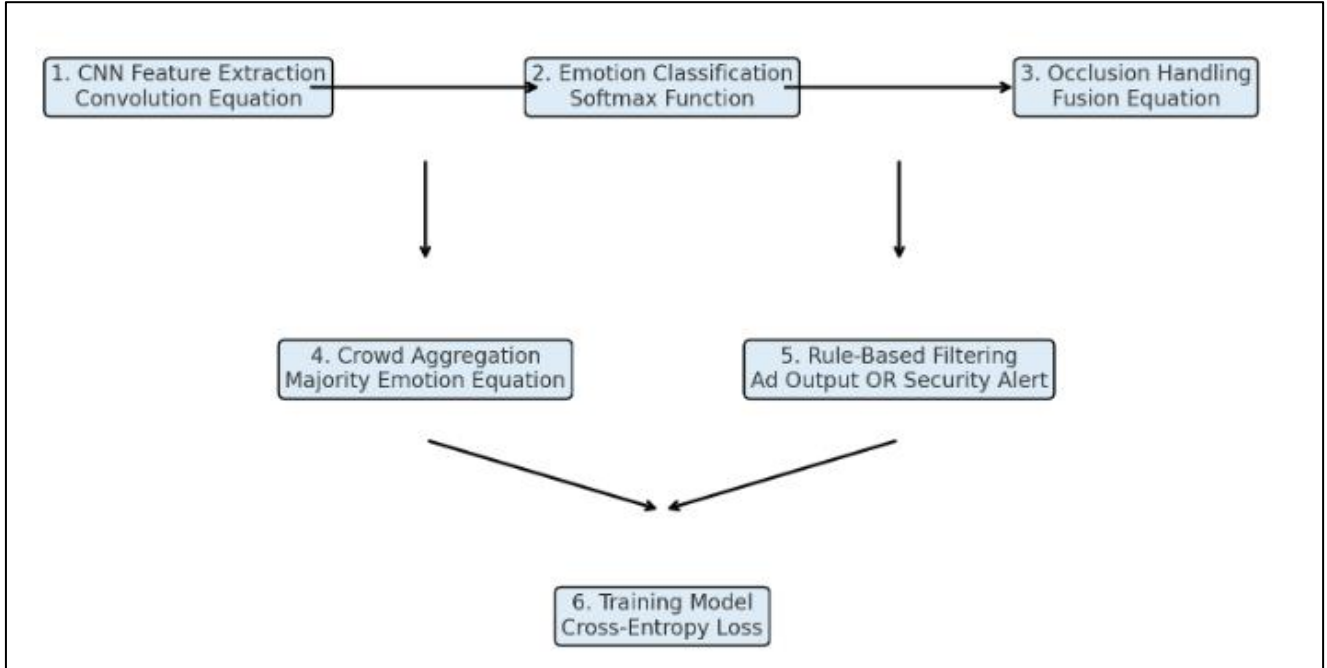


Figure 1: Mathematical Model Flow Diagram

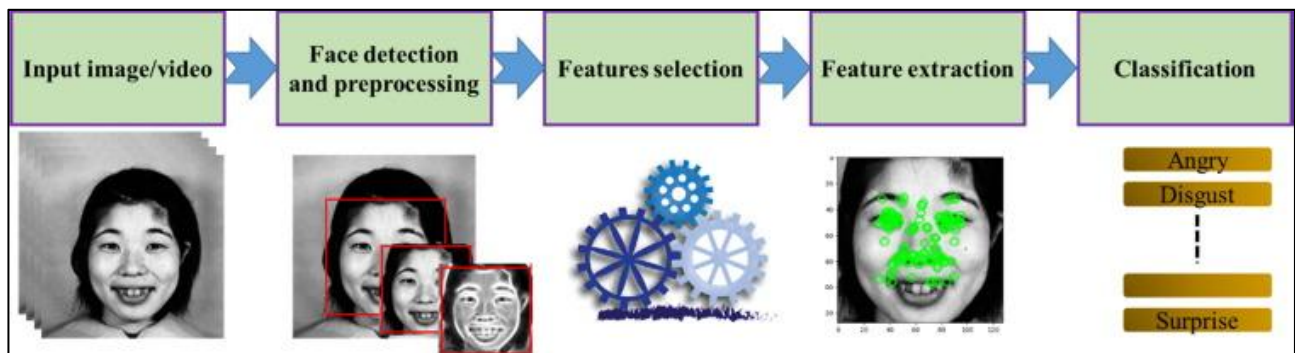


Figure 2 : Facial Expression Recognition

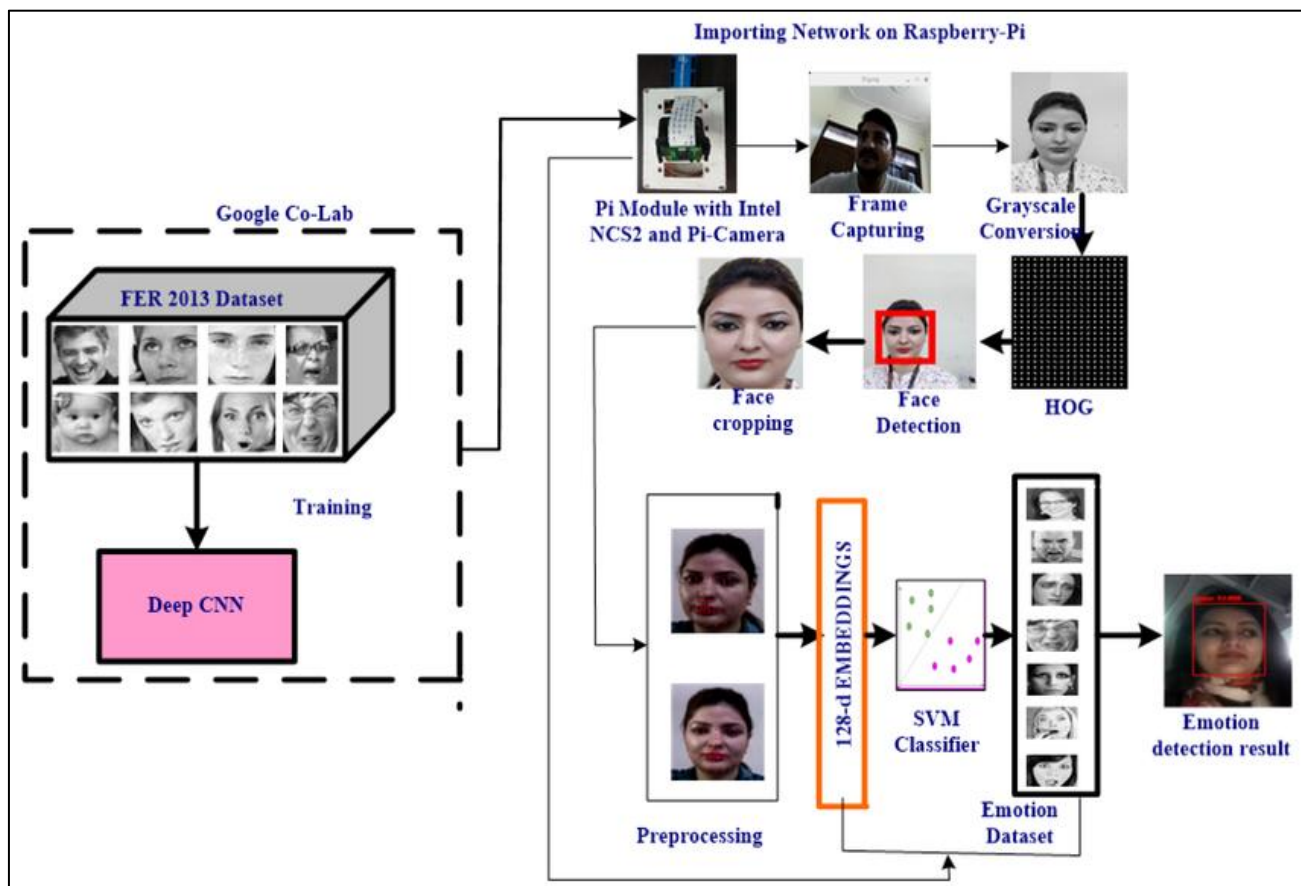


Figure 3: Raspberry Pi-based real-time emotion architecture

System Architecture Diagram

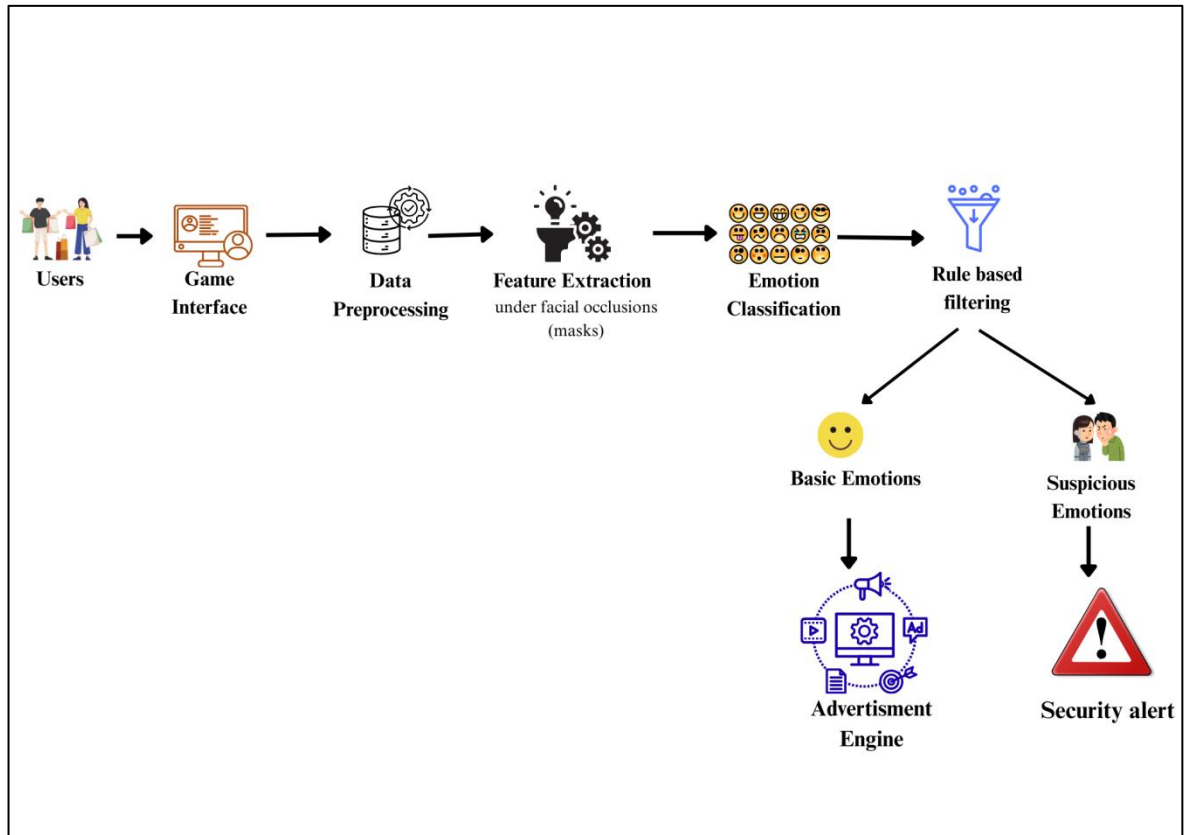


Figure 4: System Architecture Diagram of the component

The proposed system begins with users **interacting through a game interface**, which helps capture natural facial expressions. The video input then undergoes data preprocessing to improve quality before being passed into a feature extraction module, designed to handle facial occlusions such as masks or hats.

Next, a CNN-based **emotion classification model** detects **basic emotions** (happy, sad, angry, etc.) and **domain specific emotions** (suspicious, curious, ignoring). A **rule based filtering** layer then directs the outputs into two distinct pathways:

- Basic emotions are sent to the advertisement engine, which adapts digital ads in real time based on crowd mood.
- Suspicious emotions are flagged and generate an immediate security alert for mall staff.

By combining adaptive advertising and proactive security in one unified pipeline, the system enhances both customer engagement and public safety while preserving privacy through edge deployment.

Proposed Project Execution

The project will be carried out in the following phases;

1. Data Collection and Preparation:
 - Collect publicly available facial image datasets for training the emotion detection model, like Masked dataset, Sri Lankan Facial Expression dataset.
 - Preprocess and clarify data to ensure consistent labeling for basic and domain specific emotions.
 - Capture real time video footage in shopping malls for testing and ensuring the system is validated under practical conditions with partial occlusions.
2. Model Development:
 - Emotion Detection: Implement deep learning models (e.g. CNNs) to detect and classify basic and domain specific emotions. Models will be trained on labeled datasets and then fine tuned using transfer learning techniques for improved performance.
 - Occlusion Handling: Incorporate eye region and head pose features to maintain detection accuracy in scenarios with the facial occlusions.
3. Dual Output Design:
 - Develop logic to separate detected emotions into two practical outputs:
 1. Combined crowd emotion metadata for adaptive advertisements.
 2. Security alerts are triggered by repeated suspicious emotional patterns..
4. System Integration:
 - Deploy the trained model on edge devices to process live video stream in real time.
 - Integrate outputs with advertisement systems and security monitoring platform and ensuring seamless real time functionality while preserving privacy..
5. Testing and Validation:
 - Testing the system with real time video data to evaluate emotion detection accuracy under occlusion, lighting variations and crowd dynamics.
 - Collect feedback from mall operators and security teams to iteratively improve model performance and system responsiveness.

Tasks and Sub tasks Identified to Achieve the Objectives

- Task 1: Data Collection
 - Collect labeled datasets for model training.
 - Capture the real time test data from Shopping malls.
- Task 2: Model Development
 - Train and fine tune the emotion detection models.
 - Implement the occlusion handling modules.
- Task 3: System Integration
 - Implement real time emotion detection on edge devices.
 - Combine dual outputs for advertisements and security alerts.
- Task 4: Testing and Evaluation
 - Validate the system accuracy with live data.
 - Collect operational feedback and repeat improvements.

Data Requirements and Collection Methods

- Data Needed:
 - Facial Image Data: Labeled dataset containing images remark with basic and domain specific emotions. Freely available datasets include:
 - Sri Lankan Facial Emotion Dataset: 931 images (split 70% train, 15% validation, 15% test; cross validation adjustment possible).
 - MaskedFER : For training the models robust to occlusion.
 - Real Time Behavioral Data: Live video stream from mall environments for testing and validation. This include varied lighting, crowd density and partial occlusions (masks).
- Data Collection Methods:
 - Publicly available datasets will be used for the initial model training and validation.
 - Real time data will be collected from cameras to evaluate model performance in practical scenarios.
 - Data processing occurs locally on edge devices, no personally identifiable information is stored, ensuring privacy compliance.

Anticipated Conclusion and Real-World Application

Working Prototype:

The system will process live video feeds from cameras in real time, detecting emotions like happy, sad, angry, neutral, curious, suspicious and ignoring.

It will handle common occlusions like masks, which are prevalent in public spaces, by focusing on burly facial and contextual cues.

Edge device deployment ensure fast processing without relying on cloud computing, maintaining low latency and operational efficiency.

Dual Output System:

Advertising Output: Collects the detected emotional states of crowds to create a live “mood map” which can guide the selection of advertisements that match the current audience’s emotional state.

Security Output: Flags individuals or patterns that indicate suspicious / ignoring behaviors. Alerts designed to minimize false positives to avoid unnecessary engagement, providing security teams with actionable insights.

Deployment Guidelines:

Completed documentation will guide mall operators on hardware setup, edge device specifications, privacy thresholds, and best practices for system operation..

Ensures smooth integration with existing digital display and security infrastructure while maintaining user privacy by avoiding storing personal images.

Real-World Application

Adaptive Advertisement Displays:

Digital signs will dynamically change content according to collected emotional data from the crowd. [19] For example; happy or engaged crowds may trigger promotional content, while bored or neutral audiences could see more attention grabbing advertisements. This increases viewers engagement and boosts the effectiveness of marketing campaigns.

Enhanced Security Monitoring:

Detect domain specific emotions like suspicious or ignoring behaviors that allows security teams to respond proactively before potential incidents escalate. [20]

Continuously monitoring improves the situational awareness in the mall without depending entirely on human surveillance.

Privacy-Preserving Edge Deployment:

All processing occurs locally on edge devices that ensuring no personal images are stored. This design satisfies regulatory requirements and maintains visitors privacy while providing practicable insights for advertising and security. [21]

Commercialization Aspects of the product

The commercialization potential of the proposed system is significant and particularly given the growing interest in data driven retail solutions and smart mall technologies in Sri Lanka. By focusing on a place within the broader retail analytics and security market specifically, real time emotion based advertising and security alerting this system offers an unique value proposition that resonates with mall operators seeking to enhance customer engagement and safety.

This system delivers dual benefits to mall operators:

- **Enhanced Advertisement Engagement:** Digital advertisement adapt in real time to the crowd's emotions and increasing viewer interest and marketing impact.
- **Strategic Security Monitoring:** Early detection of suspicious or ignoring behaviors and allows timely intervention, improving mall safety.
- **Privacy Focused Design:** Edge processing avoids saving personal images, reducing regulatory concerns and easing adoption.

Target Market:

- **Primary Users:** Shopping mall operators, retail chains and digital display providers.
- **Secondary Users:** Security service providers pursuing intelligent alert systems.

- Psychographics: Businesses aiming for technology driven customer engagement and also enhanced security operations.

Market Space:

- Retail Analytics Market: Expanding demand for AI driven insights into customer behavior and engagement..
- Edge AI Deployment Market: High acceptance of privacy preserving AI solutions in commercial spaces.

Potential Revenue Streams:

- Software-as-a-Service (SaaS): Service plan model for the adaptive advertisement modules.
- Licensing: Security alerting functionality for the deployment in retail environments.
- Deployment & Maintenance Contracts: Hardware setup, edge device monitoring and regular model updates.

Unique Selling Proposition (USP):

- Dual Purpose System: Combines advertisement optimization and security alerts in a one solution.
- Occlusion Robust Real Time Detection: Works even with masks or partial face visibility.
- Privacy-Preserving Edge Deployment: Ensures compliance with data privacy regulations.
- Localized Optimization: Customized for Sri Lankan shopping malls using culturally relevant datasets.

Work Breakdown Structure

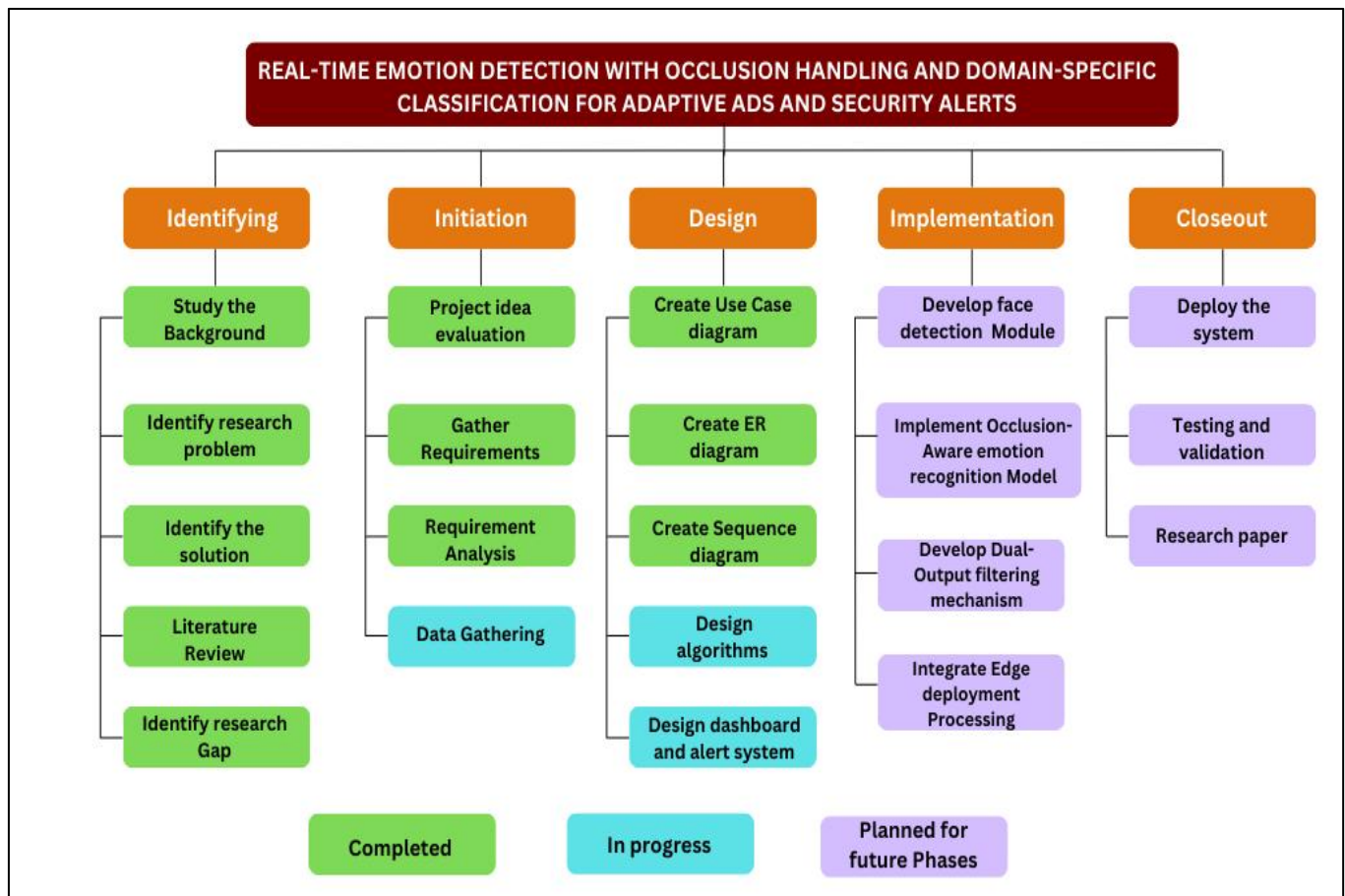


Figure 5: Work breakdown chart

Gantt Chart

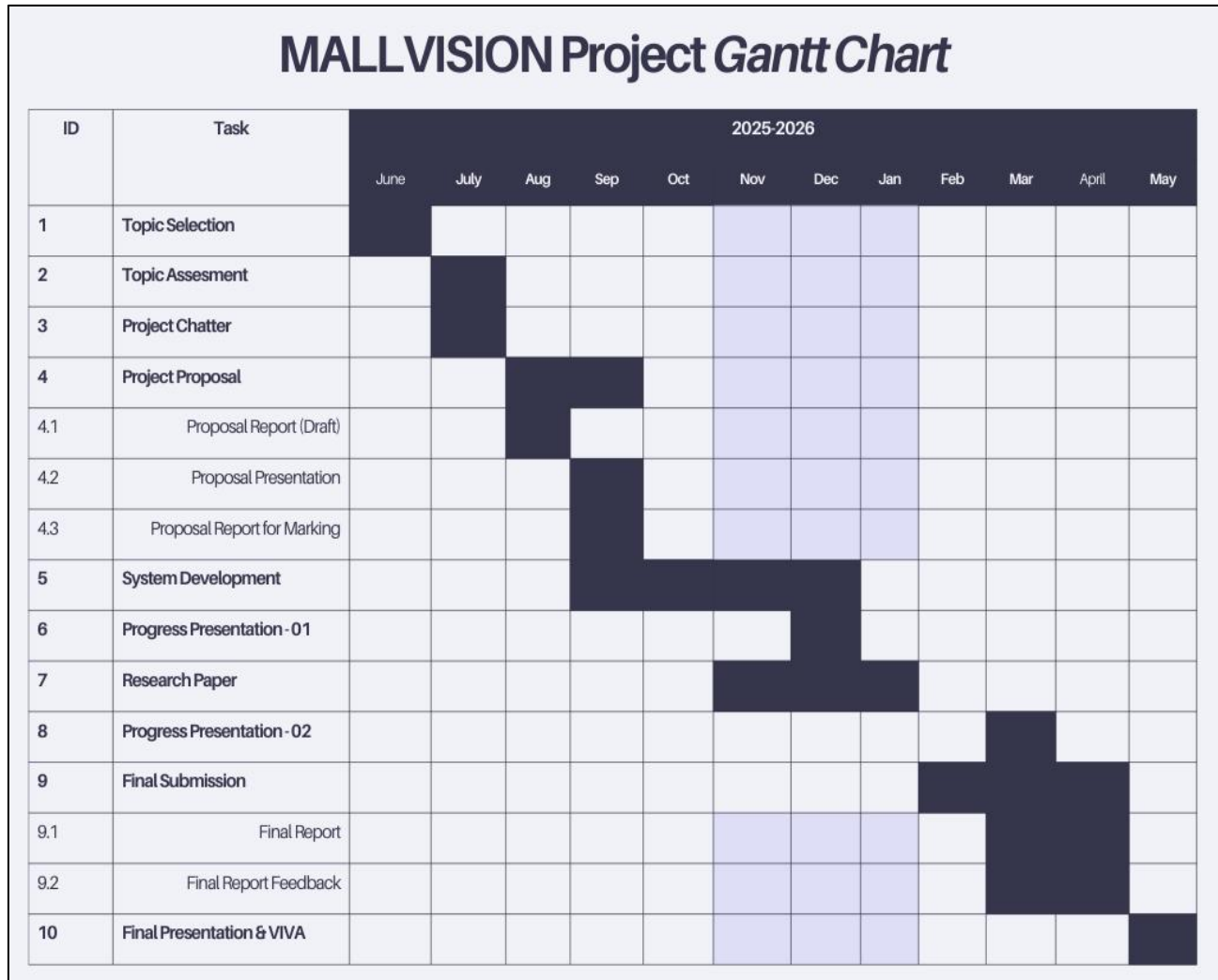


Figure 6:Gantt Chart

Use Case Diagram

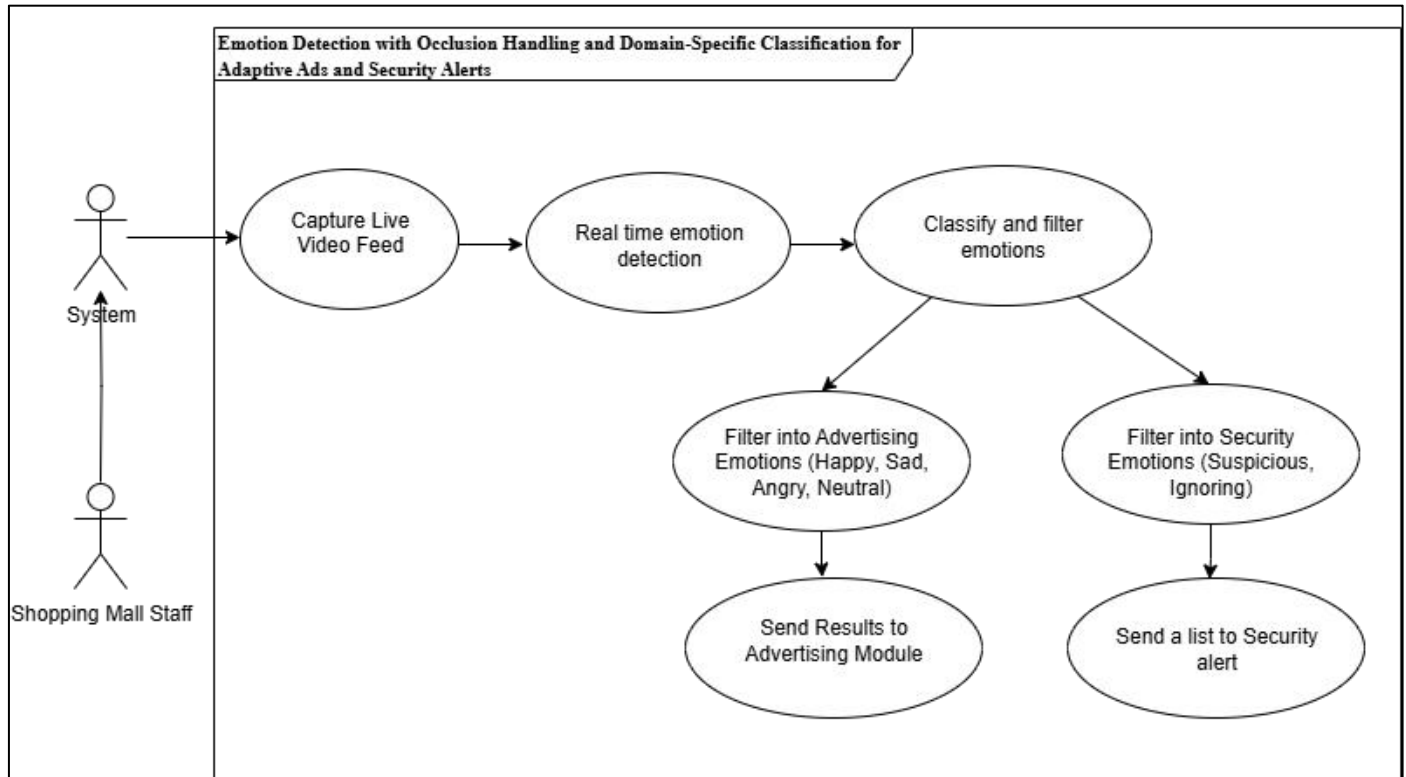


Figure 7: Use case Diagram of the component

Description of Personal and Facilities

Facilitators

- Dr. Bhagya Nathali Silva - Sri Lanka Insitute of Information Technology (SLIIT)
- Ms. Shashini Kumarasinghe - Sri Lanka Insitute of Information Technology (SLIIT)

Member Contribution

The contributions to this project involve a comprehensive approach to developing a privacy preserving, Dynamic advertising system that dynamically adapts content in shopping malls based on real time audience demographics, emotional state and behavioral analytics. Fonseka W S M focuses on audience demographic detection using computer vision models to estimate age groups in real time while ensuring that no identifiable facial data is stored, thus maintaining privacy compliance. Samaraweera W D U I works in emotion detection using computer vision models to identify emotions such as happy, sad, angry and neutral in real time and detect suspicious or ignoring behaviors for security alerts also handling occlusions to maintain accurate detection. Abeykoon S N works on pose based engagement detection and behavioral anomaly analysis, calculating the Engagement Index from body orientation and dwell time, and also identifying unusual behaviors like loitering or erratic movements for both marketing insights and security alerts. Rajana H T R develops dynamic advertisement recommendation engine, integrating demographic, emotional and behavioral data into a single pipeline that updates mall display content in real time to increase engagement.

Project requirements

System Requirements

Functional requirements

- **Real-time Video Capture & Analysis**

The system must continuously capture video streams from mall cameras and process them instantly. This ensures that both advertisements and security alerts are based on up-to-date information without delay.

- **Emotion Detection & Classification**

The model should accurately classify not only basic emotions like happy, sad, and neutral, but also domain-specific emotions such as suspicious and ignoring, which are critical for security and marketing effectiveness.

- **Occlusion Handling**

The system must remain effective even when users wear masks, caps, or sunglasses. Occlusion-aware features (eye region, head pose) will help maintain detection accuracy under real-world conditions.

- **Suspicious Behavior Alerts**

When suspicious emotions or behaviors are detected repeatedly, the system should flag them and send alerts enriched with metadata (time, location, detected state) for security monitoring.

- **Privacy by Design**

To ensure compliance with privacy regulations, the system should not store raw images or personally identifiable information. Only anonymized emotion data should be retained for analysis.

- **Continuous Monitoring**

The system should monitor emotions and crowd mood in real time, updating outputs dynamically to adapt advertisements and provide ongoing surveillance.

- **Actionable Outputs**

Detected emotions must directly connect to actionable outcomes: advertising engines should adapt content based on mood, while suspicious detections should generate security alerts.

Non Functional Requirements

- **Data Security & Encryption**

All sensitive data, including facial features and emotional states, must be protected with strong encryption methods. This prevents misuse of data and ensures compliance with data protection laws.

- **Secure Communication**

Metadata and alerts transmitted to security dashboards or ad engines must use encrypted communication channels to avoid interception or tampering.

- **Accuracy & Reliability**

The system should consistently achieve high accuracy for both occluded and non-occluded faces, with minimal false positives and negatives, ensuring reliability in practical environments.

- **Edge Deployment**

The system must run on edge devices to ensure low latency and privacy compliance. This reduces dependency on the cloud, lowers operational costs, and ensures scalability in mall settings.

User Requirements

- **Real-Time Mood Detection**

The system should continuously analyze live video feeds to detect the overall mood of the audience in real time. This ensures that advertisements and security insights are always based on the latest crowd state.

- **Dynamic Advertisement Selection**

Based on aggregated mood and demographic profiles, the system should automatically select and display advertisements that are most relevant to the audience. This personalization improves engagement and ad effectiveness.

- **Advertisement Database Management**

Mall operators should have access to a configurable advertisement database. They must be able to categorize ads, set priorities, and update content easily to align with marketing strategies.

- **Intuitive Operator Interface**

The system should provide a clear and user-friendly interface that allows mall operators to monitor real-time analytics, track mood trends, and evaluate advertisement performance without requiring advanced technical knowledge.

- **Privacy Preservation**

The system must avoid storing personal images or identifiable information. All decisions should be based only on aggregated emotion and demographic data, ensuring privacy by design while still delivering useful insights

Expected test cases (Tentative)

Table 2: Expected test cases

Test Area	Test Case ID	Description	Input Data	Expected Output
System Initialization & Video Capture	TC-1.1	Verify system starts and activates camera correctly	User starts the system	Camera activates, live feed displayed
	TC-1.2	Validate continuous video capture in real time	Ongoing camera input	Smooth video stream without frame drops
	TC-1.3	Check response when camera not available	Camera disconnected or blocked	System shows error and exits gracefully
Face Detection & Quality	TC-2.1	Verify face detection at game/camera start	User face in frame	Face detected and cropped
	TC-2.2	Validate lighting detection	Dim / bright environments	System warns if lighting insufficient
	TC-2.3	Test face positioning guidance	User off center	Prompt to reposition face shown
	TC-2.4	Verify frame capture stability	Continuous video	High quality frames processed
Emotion Detection Accuracy	TC-3.1	Test accuracy on unmasked Sri Lankan dataset	Unmasked face images	Correct classification of basic + domain-specific emotions
	TC-3.2	Test occlusion handling on masked dataset	Masked face dataset	System maintains accuracy with masks/caps

	TC-3.3	Test classification of domain specific emotions	Faces showing “suspicious/curious/ignoring”	Correct detection of domain-specific states
Security Alerts	TC-4.1	Validate suspicious detection triggers alert	Suspicious face input	Security alert generated with metadata
	TC-4.2	Test false positive handling	Normal face labeled suspicious once	No alert unless repeated
	TC-4.3	Verify alert details	Suspicious behavior detected	Alert contains timestamp, location, and metadata
Dual Output Functionality	TC-5.1	Validate ad adaptation from mood detection	Audience with happy mood	Ads updated to match mood
	TC-5.2	Validate simultaneous outputs	Crowd with mixed moods + suspicious faces	Ads adapt + alerts triggered simultaneously
Data Handling & Privacy	TC-6.1	Verify no raw image storage	Raw face input	Images deleted after processing
	TC-6.2	Test secure metadata transmission	Detection results	Metadata encrypted and sent securely

Budget and Budget Justification

The projected budget primarily consists of expenses for hardware required to develop the wearable device, along with travel related costs. Since our focus is on the research outcome as a complete system, we do not intend to commercialize the individual components separately. Instead, final product will be presented as an integrated solution.

Table 3: Budget Estimation

Purpose	Cost (LKR)
Hardware Cost	20000
Traveling Cost	5000
Software and Subscription Cost	6000
Total	31000

References

- [1] W. L. J. H.-w. H. Jackie Wai Yi Wo, "Impact of mask use on facial emotion recognition in individuals with subclinical social anxiety: an eye-tracking study," 02 06 2025.
- [2] C. W. S. B. T. D. W. G. Kangning Yang, "Benchmarking commercial emotion detection systems using realistic," *The Visual Computer*, 2020.
- [3] P. Britt, "Emotion Detection in Marketing Is About to Go Mainstream," CRM, 2023.
- [4] F. C. Martina Mattioli, "Not in My Face: Challenges and Ethical Considerations in Automatic Face Emotion Recognition Technology," *MDPI*, 2024.
- [5] D. R. Amod Pathirana, "Facial Expression Dataset (Sri Lanka)," IEEE, 2025.
- [6] J. K. a. D. Lee, "Facial Expression Recognition Robust to Occlusion and to Intra-Similarity Problem Using Relevant Subsampling," 27 2 2023.
- [7] J. M. A. A. R. A. L. Ronak Kosti, "Context Based Emotion Recognition using," *IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE*.
- [8] H. L. J. L. a. D. J. Yunfei Li, "Occlusion-Robust Facial Expression Recognition Based on Multi-Angle Feature Extraction," *MDPI*, 2025.
- [9] N. V. Thanh, "Emotion Recognition Systems in Retail: A Detailed Analysis of Their Role in Enhancing Customer Interactions, Driving Sales, and Predicting Trends," *Journal of Computational Socail Dynamics*, 2023.
- [10] E. C. M. P. E. C. V. J.-i. K. J.-W. J. Y. J. S.-H. K. a. W. L. Alexander M. Pascual 1ORCID, "Light-FER: A Lightweight Facial Emotion Recognition System on Edge Devices," *MDPI*, 2022.
- [11] M. A. P. I. A. M. V. G. B. Mike Rinck, "Face masks impair facial emotion recognition and induce specific emotion confusions," *Cognitive Psychology*, 2022.
- [12] B. V. D. T. V. C. Ligang Zhang, "Facial Expression Analysis under Partial Occlusion: A Survey," *Computer Vision and Pattern Recognition*, 2018.
- [13] Wikipedia, "Emotions and culture," 5 2012.
- [14] F. Soft, "7 Leading Real-Time AI Emotion Recognition Software Solutions," Hong Kong, 2025.
- [15] M. A. P. I. A. M. V. & G. B. Mike Rinck, "Face masks impair facial emotion recognition and induce specific emotion confusions," *Cognitive Research: Principles and Implications*, 2022.
- [16] E. R. A. G. 2. C. Vlad Pandelea, "Emotion Recognition on Edge Devices: Training and Deployment," 2021.
- [17] A. E. & M. A. A. Yasmee ELsayed, "An automatic improved facial expression recognition for masked faces," *Neural Computing and Applications*, 2023.
- [18] M. Burgess, "Amazon-Powered AI Cameras Used to Detect Emotions of Unwitting UK Train Passengers," *Wired - Security*, 2024.
- [19] S. B. Tarun Gupta, "AI-Driven Emotional Recognition in Digital Ads: A Novel Approach to Consumer Engagement," *Journal of Marketing & Supply Chain Management*, 2023.
- [20] G. K. Amol Patwardhan, "Aggressive actions and anger detection from multiple modalities using Kinect," *Human-Computer Interaction*, 2016.
- [21] F. u. H. S. L. H. U. M. J. Q. Hafiz Shehbaz Ali, "Privacy Enhanced Speech Emotion Communication using Deep Learning Aided Edge Computing," 31 3 2021.